Practice workbook · Draw a line or curve of best fit through plotted points

Points are already plotted and a single straight line or smooth curve showing the trend is requested.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-LINE-OF-BEST-FIT

1. Question 2 · part a.ii · 1 mark

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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1. Question 2 · part a.ii · 1 mark

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

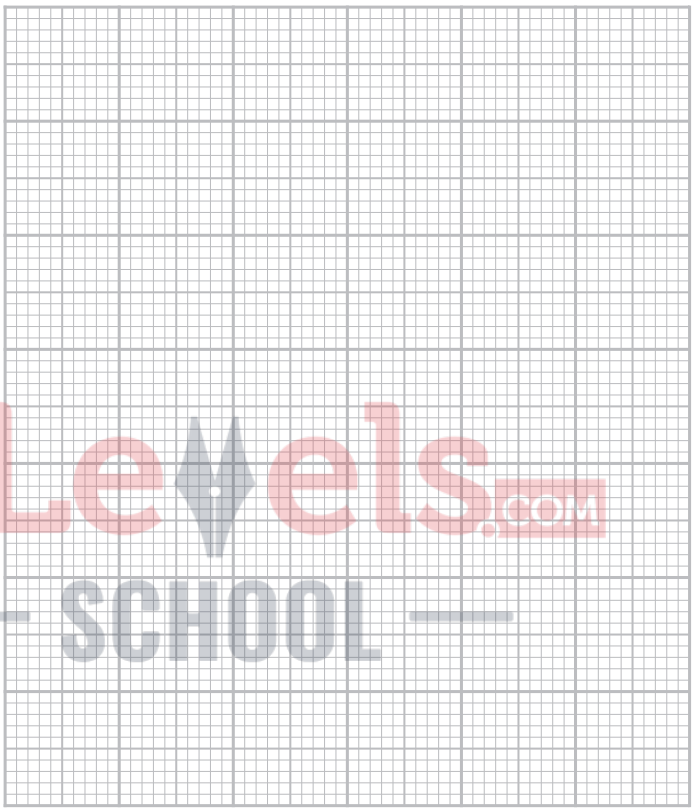
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

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1. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

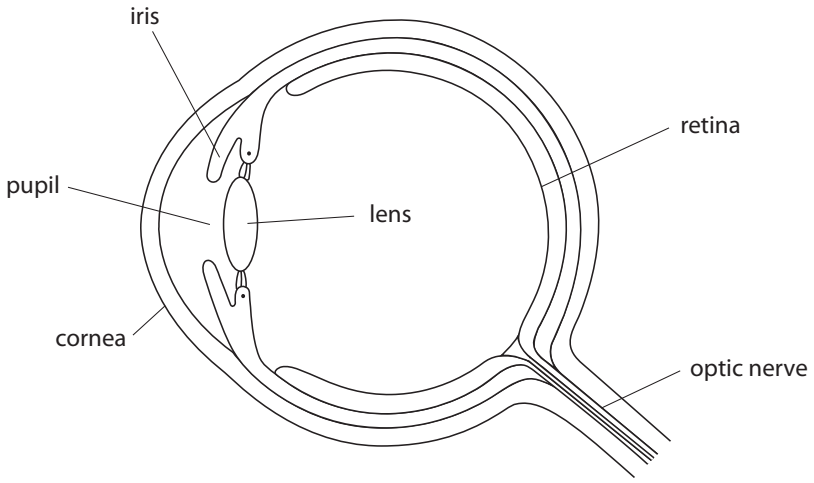
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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

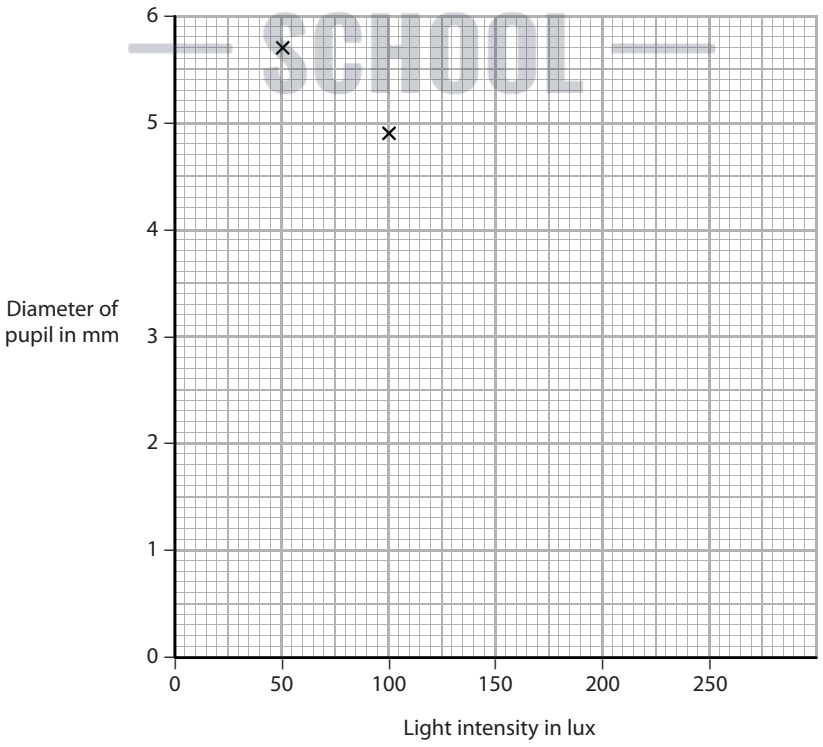
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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Working and answer

3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

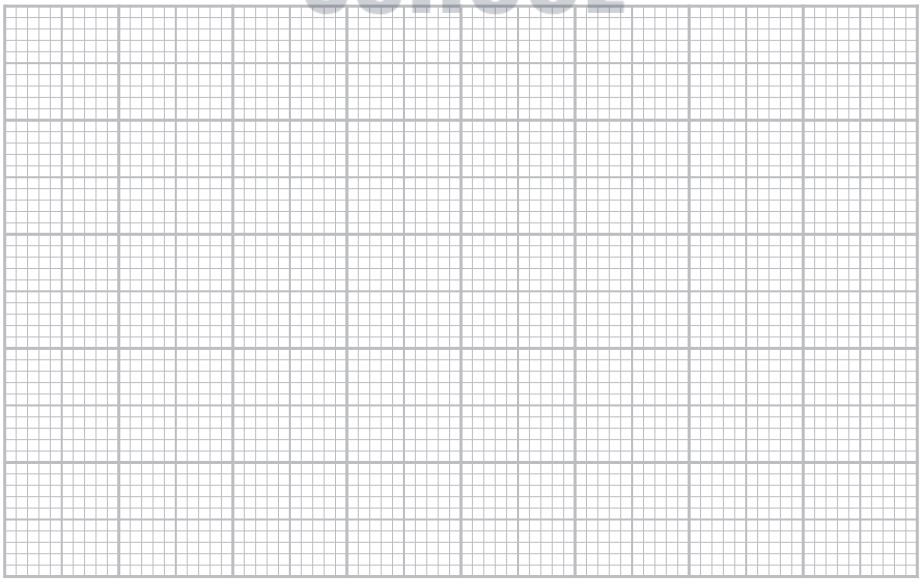
Further practice · Attempt part b.ii only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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(Total for Question 2 = 13 marks)

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Working and answer

4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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5 Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

(2)

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

(2)

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4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

Period of time in hours	Percentage change in mass of haemoglobin
0	0
100	0.7
200	2.4
300	4.1
400	5.9
500	6.5
600	7.2

(i) Plot the results on the grid.

(4)

(ii) Draw the curve of best fit.

(1)

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4. Question 5 · part b.ii · 1 mark

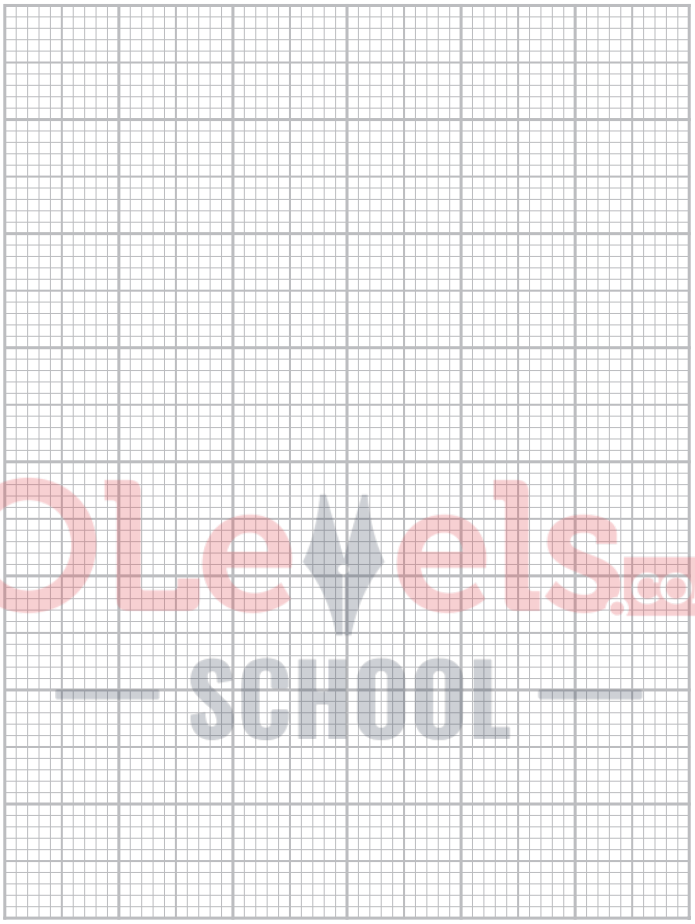
January 2023 Paper 2

Further practice · Attempt part b.ii only.

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4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(iii) Discuss how the percentage change in mass of haemoglobin, after training at altitude, could result in an increase in performance during exercise.

(5)

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(Total for Question 5 = 14 marks)

4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 4 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

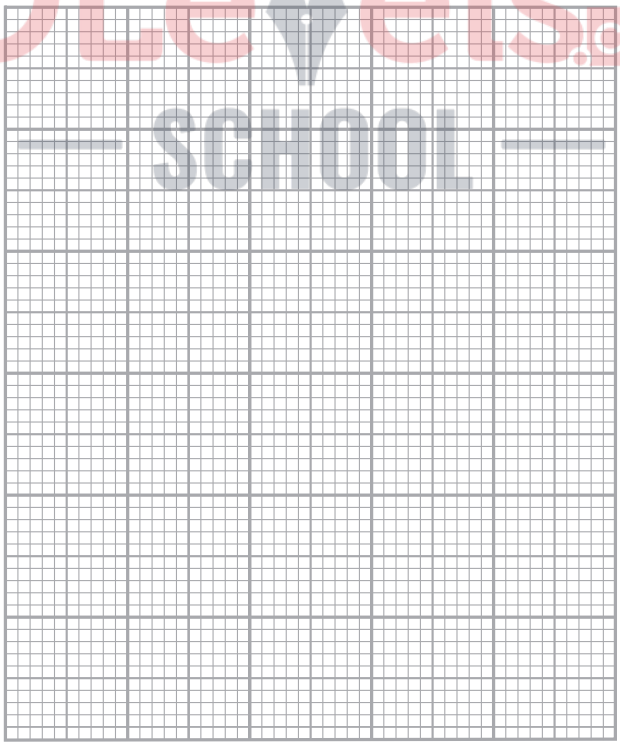
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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5. Question 4 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

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5. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 4 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

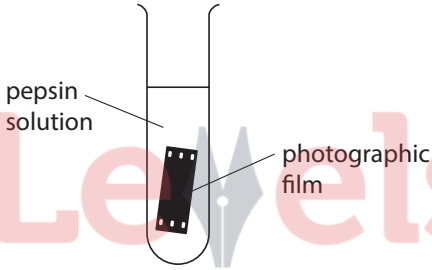
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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6. Question 4 · part a.ii · 1 mark

June 2025 Paper 1R

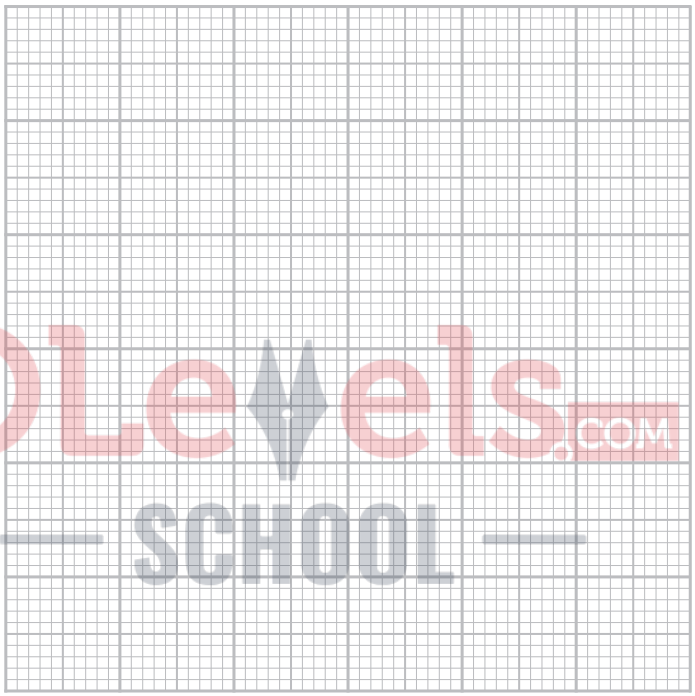
Further practice · Attempt part a.ii only.

(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



(b) (i) Explain why the photographic film becomes clear.

(3)

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6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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